

RECENT DEVELOPMENTS IN THE LANGLANDS PROGRAM

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ABSTRACT. Cette article est une exposition des résultats récents dans le programme de Langlands. Notamment, c'est un exposé du travail de Shahidi et Kim sur la théorie des fonctions L attachées aux puissances symétriques tensorielles d'une représentation automorphe de GL_n .

This article is based on a series of lectures delivered at the Centre de Recherches Mathématiques in Montréal in April 2002. These lectures focussed solely on recent progress in the Langlands program over number fields. Spectacular progress has been made recently in the function field case but we do not report on that here.

We will begin with an informal motivation discussing four outstanding conjectures (still unresolved) that form the background thought and subconscious force that drives much of the program. These are the Sato-Tate conjecture, the Ramanujan-Petersson conjecture, the Selberg eigenvalue conjecture, and Artin's conjecture about non-abelian L -functions attached to complex linear Galois representations. Then, we will give a brief and formal description of the construction of Langlands' L -functions and state the basic conjectures about them. In the final sections, we report on the progress made so far towards these conjectures highlighting those advances made in the last decade.

1. The Sato-Tate conjecture.

Given an elliptic curve E over a number field F , we know that for each prime ideal v of F where E has good reduction, the number of points of $E \bmod v$ is given by

$$N(v) + 1 - a_v$$

where $N(v)$ denotes the norm of v and a_v satisfies Hasse's inequality

$$|a_v| \leq 2(N(v))^{1/2}.$$

One can therefore write

$$a_v = 2N(v)^{1/2} \cos \theta_v$$

where $0 \leq \theta_v \leq \pi$. The Sato-Tate conjecture is a statement about how the angles θ_v are distributed in the interval $[0, \pi]$ as v varies. In the case that E has complex multiplication, the behaviour is known and is a classical result of Hecke [H, p. 215-234, 249-281]. In particular, Hecke's theorem says that if the CM field is contained in F , then the angles are uniformly distributed in $[0, \pi]$ in the sense of Weyl.

By contrast, in the non-CM case, Tate [T] and independently Sato conjectured that the angles are not uniformly distributed with respect to the usual Lebesgue measure but rather with respect to the measure

$$\frac{2}{\pi} \sin^2 \theta d\theta.$$

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More precisely, they conjecture that the number of prime ideals v with norm less than x with $\theta_v \in (\alpha, \beta)$ is asymptotic as x tends to infinity, to

$$\left(\frac{2}{\pi} \int_{\alpha}^{\beta} \sin^2 \theta d\theta \right) \pi_F(x),$$

with $\pi_F(x)$ denoting the number of prime ideals with norm less than or equal to x . In a recent paper, Akiyama and Tanigawa [AT] make the stronger conjecture that the error term in this asymptotic formula is

$$O(x^{1/2+\epsilon})$$

for any $\epsilon > 0$. By applying Koksma's inequality in the theory of Diophantine approximation, they show that their conjecture implies that $L_1(s)$ (defined below) satisfies the analog of the Riemann hypothesis. A minor modification of their proof shows that in fact, all of the $L_m(s)$ (see below) satisfy the Riemann hypothesis as a consequence of their strong form of the Sato-Tate conjecture.

In the special case that E is defined over the rational number field, the modularity conjecture of Taniyama, Shimura and Weil (now a theorem) says that there exists a cusp form f_E of weight 2 (see definition in section 2) such that for each prime p of good reduction for E , a_p is the p -th Fourier coefficient $a_p(f_E)$ of f_E . It then seems plausible that the Sato-Tate conjecture is really a statement about Fourier coefficients of cusp forms which are Hecke eigenforms. That this indeed should be the case, is predicted by Serre [Se]. More precisely, if f is a new form of weight k and level N , we may write

$$a_p(f) = 2p^{(k-1)/2} \cos \theta_p.$$

If f is not of CM-type, then Serre predicts that the angles θ_p are distributed with respect to the Sato-Tate measure

$$\frac{2}{\pi} \sin^2 \theta d\theta$$

as in the elliptic curve case.

In a fundamental paper, Langlands [La] outlined an approach to this conjecture via the theory of L -functions. We will be more precise about the origins of these L -functions in a later section. For now, they can be defined as follows.

Let $\alpha_p = e^{i\theta_p}$ and $\beta_p = e^{-i\theta_p}$. For each natural number m , set

$$L_m(s, f) = \prod_{(p, N)=1} \prod_{j=0}^m \left(1 - \frac{\alpha_p^{m-j} \beta_p^j}{p^s} \right)^{-1}.$$

These L -functions are referred to as symmetric power L -functions. Clearly, $L_m(s, f)$ converges absolutely for $\Re(s) > 1$. For $m = 0$, we have $L_0(s, f)$ is (apart from a few Euler factors) equal to the Riemann zeta function. Langlands [La] indicated that if each $L_m(s, f)$ $m \geq 1$, extends to an analytic function and does not vanish on the line $\Re(s) = 1$, then

the Sato-Tate conjecture follows as a consequence. A more rigorous treatment and proof was later provided by Serre [Se]. Subsequently, Kumar Murty [VKM] showed that the non-vanishing hypothesis can be dropped since the analytic continuation and a superb generalization of the classical technique of Hadamard and de la Vallée Poussin establish that $L_m(1+it, f) \neq 0$. In this way, the Sato-Tate conjecture has been reduced to a problem of analytic continuation of certain L -functions.

Concerning the functions $L_m(s, f)$ for $m \geq 1$, Langlands conjectures that each of them extends to an entire function of s . This problem lies at the heart of the Langlands program and has been the muse of many discoveries and advances in the theory.

What is known about these analytic continuations? For $m = 1$, the assertion that $L_1(s, f)$ extends to an entire function is the work of Hecke [H]. It comes as a consequence of the theory of modular forms. Thus, encapsulated in $L_1(s, f)$ is a good chunk of what is called GL_2 -theory which we will elaborate on below.

The method of Rankin [Ra] and Selberg [Sel] shows that

$$\zeta(s)L_2(s, f)$$

extends to $\Re(s) = 1$ except for a simple pole at $s = 1$. As the Riemann zeta function does not vanish on $\Re(s) = 1$ $s \neq 1$, and at $s = 1$, it has only a simple pole, we conclude that $L_2(s, f)$ extends to an analytic function for $\Re(s) \geq 1$. The Rankin-Selberg method however, does not allow us to conclude that $L_2(s, f)$ itself extends to an entire function since it does not exclude the possibility that $L_2(s, f)$ has poles at the zeros of $\zeta(s)$. Thus a new technique was needed to deduce the required property of $L_2(s, f)$. This is a famous theorem of Shimura [Sh] which signalled the famous Shimura correspondence between cusp forms of integral weight and certain cusp forms of half-integral weight.

Recently Shahidi and Kim [KS] have proved:

Theorem 1. $L_3(s, f)$ extends to an entire function.

Very recently, Kim [K] has proved that

Theorem 2. $L_4(s, f)$ extends to an entire function.

As a further consequence of their work, Kim and Shahidi [KS] have established that for $5 \leq m \leq 9$, $L_m(s)$ has meromorphic continuation to the entire complex plane.

These results represent tremendous progress in the Langlands program and have significant consequences to other questions we will consider, namely to the Ramanujan-Petersson conjecture and the Selberg eigenvalue conjecture which we will elucidate below.

2. Maass forms.

Let $\mathcal{H}$ be the upper half plane. The group $SL_2(\mathbb{Z})$ acts on the upper half-plane via the fractional linear transformations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}.$$

It is convenient to consider the extended upper half-plane $\mathcal{H}^*$ which is $\mathcal{H}$ together with all the rational numbers and the point $i\infty$. These additional points (that is the set of rational numbers and $i\infty$) are called cusps. One extends the action of $SL_2(\mathbb{Z})$ to $\mathcal{H}^*$ in the obvious way. Let $\Gamma_0(N)$ be the (Hecke) congruence subgroup of $SL_2(\mathbb{Z})$ defined by the set

$$\left\{ \gamma \in SL_2(\mathbb{Z}) : \gamma \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}.$$

Let χ be a Dirichlet character mod N . A modular form of weight k and level N and Nebentypus χ is defined as a holomorphic function $f : \mathcal{H}^* \rightarrow \mathbb{C}$ satisfying the transformation

$$f\left(\frac{az+b}{cz+d}\right) = \chi(d)(cz+d)^k f(z), \quad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N).$$

The set of all such forms comprises a finite dimensional $\mathbb{C}$ -vector space, denoted $M_k(N, \chi)$ over the complex numbers. The subspace of forms vanishing at all the cusps is denoted $S_k(N, \chi)$. The elements of this space are called cusp forms. Since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \Gamma_0(N),$$

the condition above implies that any modular form f has a Fourier expansion of the form

$$f(z) = \sum_{n=0}^{\infty} a_n(f) e^{2\pi i n z}.$$

Any such function $f(z)$ can also be viewed as a smooth function $f(x, y)$ of two real variables x, y and the holomorphy condition can be reformulated using the Cauchy-Riemann equations. Consequently $f(x, y)$ is an eigenfunction of the non-Euclidean Laplacian

$$\Delta = -y^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right).$$

If we drop the condition of holomorphy from the definition of a modular form and insist only that our function still is an eigenfunction of Δ , we arrive at the notion of a Maass wave form. Such forms necessarily have a Fourier expansion of the form

$$f(x, y) = \sum_{n=0}^{\infty} a_n(f, y) e^{2\pi i n x}.$$

The condition that f be an eigenfunction of the Laplacian with eigenvalue λ means that f satisfies the (Whittaker) differential equation

$$-y^2 \frac{d^2}{dy^2} a_n(f, y) = (\lambda - 4\pi^2 n^2 y^2) a_n(f, y).$$

If we put $b_n(f, y) = a_n(f, y/2\pi n)/\sqrt{y/2\pi n}$, then $b_n(f, y)$ satisfies the famous Bessel equation:

$$y^2 \frac{d^2}{dy^2} b_n(f, y) + y \frac{d}{dy} b_n(f, y) - (y^2 + (\frac{1}{4} - \lambda)) b_n(f, y) = 0.$$

The solutions of these equations have been well studied for over two centuries. For $n = 0$, Whittaker's equation simplifies to

$$y^2 \frac{d^2}{dy^2} a_0(f, y) = -\lambda a_0(f, y)$$

and it is easy to see that all solutions are of the form

$$b_0(y) = ay^s + by^{1-s}, \quad \lambda = s(1-s).$$

Observe that if $\lambda = 0$, that is, f is holomorphic, then the equation simplifies to

$$\frac{d^2}{dy^2} a_n(f, y) = 4\pi^2 n^2 a_n(f, y)$$

and any solution is of the form

$$a_n(f, y) = a_n(f) e^{-2\pi n y} + a_n(f)' e^{2\pi n y}.$$

Since our functions are holomorphic at $i\infty$, the second solution drops out and we find that our solution for $\lambda = 0$ has the form

$$f(x, y) = a_0 + \sum_{n=1}^{\infty} a_n(f) e^{2\pi i n z}$$

which is the familiar Fourier expansion of a modular form.

In the general case, when $\lambda \neq 0$, we write

$$\lambda = \frac{1}{4} + r^2$$

and derive the Fourier expansion for f as

$$f(z) = a_0(f) y^s + a_0'(f) y^{1-s} + \sum_{n \neq 0} a_n(f) \sqrt{y} K_{ir}(2\pi |n| y) e^{2\pi i n x},$$

where

$$K_{ir}(y) = \frac{1}{2} \int_{-\infty}^{\infty} e^{-y \cosh t} e^{-irt} dt.$$

Such a function is called a Maass wave form corresponding to the eigenvalue λ . If $f(z)$ is a Maass wave form, so is $\iota \circ f(z) = f(-\bar{z})$ with the same eigenvalue. As $\iota^2 = 1$, we can diagonalize the space of Maass forms with eigenvalue λ with respect to ι . If $\iota \circ f = f$, we

say f is even and if $\iota \circ f = -f$, we say f is odd. Maass proved that if f is a cusp form so that $a_0(f)$ and $a_0(f)'$ both vanish, then the Dirichlet series

$$L(s, f) = \sum_{n=1}^{\infty} \frac{a_n(f)}{n^s}$$

extends to an entire function and satisfies a functional equation of the form

$$A^s \Gamma\left(\frac{s + \delta - ir}{2}\right) \Gamma\left(\frac{s + \delta + ir}{2}\right) L(s, f) = \\ w A^{1-s} \Gamma\left(\frac{1 - s + \delta - ir}{2}\right) \Gamma\left(\frac{1 - s + \delta + ir}{2}\right) L(1 - s, \bar{f})$$

for a suitable constant A , and w of absolute value 1, and $\delta = 0$ or -1 depending on whether f is even or odd. (For further details about Maass forms, see [Bu, pp. 103-119].)

The Ramanujan-Petersson conjecture (still unresolved) predicts that

$$a_n(f) = O(n^\epsilon)$$

for any $\epsilon > 0$. The best result known as a consequence of the work of Shahidi and Kim is

$$a_n(f) = O(n^{7/64})$$

due to Kim and Sarnak [KSk].

Selberg's eigenvalue conjecture predicts that

$$\lambda \geq \frac{1}{4}.$$

Selberg himself proved it in the case $N = 1$ and showed in general that

$$\lambda \geq \frac{3}{16}.$$

The best result here, again due to Kim and Sarnak is

$$\lambda \geq \frac{975}{4096} = .238.$$

It is also an open problem to determine the dimension of the space of eigenspace corresponding to the eigenvalue λ . In the case $N = 1$, this space is conjectured to be one-dimensional.

All of these results are made possible because of our increasing knowledge of symmetric power L -functions. Indeed, in the paper alluded to above, Langlands [La] showed how one

can construct for each Maass form, a family of L -functions $L_m(s, f)$ similar to the case of holomorphic f . However, unlike the holomorphic case, it is not known at present if all of these L -functions are analytic in a fixed half-plane. If this were so, Langlands showed that both the Ramanujan-Petersson and Selberg conjectures would follow. We will describe this in some more detail in the section below.

3. Artin's conjecture.

As before, let F be an algebraic number field with ring of integers O_F . Let K/F be a finite Galois extension with group G . Given a prime ideal $\mathfrak{p}$ of F , we have by a classical theorem of Dedekind a unique factorization

$$\mathfrak{p}O_K = \wp_1^{e_1} \cdots \wp_g^{e_g}$$

into prime ideals $\wp_i$. For each prime ideal $\wp$ of K , we define the decomposition group

$$D_\wp = \{\sigma \in G : \sigma(\wp) = \wp\}$$

and a normal subgroup of $D_\wp$ called the inertia group

$$I_\wp = \{\sigma \in G : \sigma(x) \equiv x \pmod{\wp}, \quad \forall x \in \mathcal{O}_K\}.$$

The quotient group $D_\wp/I_\wp$ is canonically isomorphic to the Galois group of the finite field extension $O_K/\wp$ over $O_F/\mathfrak{p}$. It has a generator $\sigma_\wp$ called the Frobenius automorphism that is uniquely determined by the property

$$\sigma_\wp(x) \equiv x^{N_{F/\mathbb{Q}}(\mathfrak{p})} \pmod{\wp}.$$

Thus, $\sigma_\wp$ is determined uniquely modulo the inertia subgroup (which is trivial apart from a finite set of primes which are ramified in K). Its conjugacy class is denoted $\sigma_\mathfrak{p}$ and is called the Artin symbol of $\mathfrak{p}$ in the literature.

Now suppose that V is a finite dimensional representation of G . That is, we have a homomorphism

$$\rho : G \rightarrow GL(V),$$

with V a finite-dimensional vector space over $\mathbb{C}$. Artin attached an L -function to such a representation by setting:

$$L(s, \rho; K/F) = \prod_{\mathfrak{p}} \det(1 - \rho(\sigma_\mathfrak{p}) N_{F/\mathbb{Q}}(\mathfrak{p})^{-s} |V^{I_\wp}|)^{-1}$$

where as remarked above $\sigma_\wp$ is uniquely determined modulo the inertia subgroup so that the determinant in the definition above is well-defined since it only depends on the conjugacy class of $\wp$ for any $\wp|\mathfrak{p}$ and we are restricting to the subspace fixed by the inertia subgroup $I_\wp$.

These L -functions have certain functorial properties. First, it is clear that it only depends on the character χ of ρ , so therefore, we denote $L(s, \rho; K/F)$ by $L(s, \chi; K/F)$. Also, if $\chi = \chi_1 + \chi_2$, then

$$L(s, \chi; K/F) = L(s, \chi_1; K/F)L(s, \chi_2; K/F).$$

It is less clear that Artin L -series are invariant under the induction of characters. This was proved by Artin. Namely, if H is a subgroup of G and K^H is its fixed subfield, then for any character ψ of H , we have

$$L(s, \psi; K/K^H) = L(s, \text{Ind}_H^G \psi; K/F).$$

Artin conjectured that for any irreducible character χ of G which is not trivial, $L(s, \chi; K/F)$ extends to an entire function. As a culmination of the development of class field theory, he showed that his conjecture is true for any one-dimensional character. This theorem embodies the famous reciprocity law of Artin [A].

Motivated by Artin's conjecture, Brauer [B] proved a fundamental theorem in group theory now called Brauer's induction theorem. It says that given any character χ of G we may write it as a $\mathbb{Z}$ -linear combination

$$\chi = \sum_i n_i \text{Ind}_{H_i}^G \psi_i$$

with ψ_i one-dimensional. Thus,

$$L(s, \chi; K/F) = \prod_i L(s, \text{Ind}_{H_i}^G \psi_i; K/F)^{n_i}.$$

Now using the fact that Artin L -series are invariant under induction and the reciprocity law, we deduce that every Artin L -function extends to a meromorphic function of s .

One of the consequences of Artin's conjecture is Dedekind's conjecture that the Dedekind zeta function $\zeta_M(s)$ of a finite extension of F is always divisible by $\zeta_F(s)$. That is, $\zeta_M(s)/\zeta_F(s)$ is entire. If M/F is Galois, Dedekind's conjecture is known and this is the celebrated theorem of Aramata and Brauer. Dedekind's conjecture is also known to be true whenever M is contained in a solvable extension M of F . A simple proof can be found in [KM]. In the paper [KM] (see also [FM]), Kumar Murty proves by an exceedingly simple method the following result. Fix $s_0 \in \mathbb{C}$ and let n_χ be the order of the zero (or pole) $L(s, \chi, K/F)$ at $s = s_0$. Then

$$\sum_\chi n_\chi^2 \leq r_0^2$$

where the sum is over all the irreducible characters of $\text{Gal}(K/F)$ and r_0 is the order of the Dedekind zeta function $\zeta_K(s)$ at $s = s_0$. This implies the upper bound $|n_\chi| \leq r_0$ and as a consequence, we can deduce that for any irreducible character χ ,

$$L(s, \chi, K/F)\zeta_K(s)$$

is regular for all $s \neq 1$. It is actually possible to be a bit more precise. By a careful application of the Cauchy-Schwarz inequality in [KM], one can derive the sharper estimate $|n_\chi| < r_0$ whenever $n_\chi < 0$.

A primary goal of the Langlands program is to prove Artin's conjecture. In fact, certain L -functions already defined by Langlands have been shown to extend to entire functions by Godement and Jacquet [GJ]. The famous Langlands reciprocity conjecture is that each Artin L -function attached to an irreducible character is one of these functions already known to be entire.

This aspect of the program has had remarkable success. For instance, in the case of two dimensional Galois representations ρ , the following theorem is a consequence of the work of Langlands [La2] and Tunnell [Tu]:

Theorem 3. *If ρ is a two-dimensional representation of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with solvable image in $\text{PGL}_2(\mathbb{C})$, then Artin's conjecture is true.*

In fact, more is true. The Artin L -function is automorphic (in the sense to be defined below). This theorem, can be viewed as a two-dimensional analogue of Artin's reciprocity law. This theorem, certainly being one of the pinnacles of 20th century mathematics was the starting point of the proof of Wiles [Wi] of the modularity conjecture referred to above. Combining this work with that of Ribet, one obtains a complete proof of the much celebrated Fermat's Last Theorem.

In case the image is non-solvable, some new results have been established, notably in the work [BDST].

4. The adèle ring $\mathbb{A}_F$.

To begin our description of the Langlands program, we need to introduce adeles and adèle groups as well as discuss their representation theory. The notion of an adèle can be motivated in the following way in the case of the rational number field. It can be regarded as an infinite tuple

$$a = (a_\infty, a_2, a_3, a_5, \dots)$$

with $a_\infty \in \mathbb{R}$ and $a_p \in \mathbb{Q}_p$, the p -adic number field, so that $a_p \in \mathbb{Z}_p$ (the ring of p -adic integers) for p sufficiently large. The set of all such adeles can be made into a ring $\mathbb{A}_\mathbb{Q}$ by componentwise addition and multiplication. Observe that the rationals can be viewed as a subring of the adeles of the adeles via the embedding

$$x \mapsto (x, x, x, \dots).$$

To each such a , we can define a character χ_a on the additive group of the rationals by setting

$$\chi_a(x) = \exp(2\pi i(-a_\infty x + a_2 x + a_3 x + \dots))$$

where we interpret the exponential as follows. For p sufficiently large, x is a p -adic integer and so $a_p x$ is a p -adic integer, so we ignore these in the evaluation of the exponential. For the remaining finitely many primes, we can write $a_p x$ as the sum of rational number

(determined from the “polar part” of $a_p x$) and a p -adic integer, which again we ignore in the evaluation of the exponential. In this way, we view the argument of the exponential as a real number and evaluate the exponential as a function of a real variable. One can easily check that this gives a well-defined character of the additive group of the rationals. Moreover, two adeles a and b give rise to the same character if and only if

$$a - b = (x, x, x, \dots)$$

for some rational number x . Thus, the character group of the rationals is isomorphic to $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$. The adelic topology, defined below, is made so that this isomorphism is one of topological groups.

Generally, for an algebraic number field F , we define the adèle ring $\mathbb{A}_F$ as follows. For each place v of F let F_v be the completion of F at v . Thus, in the case v is finite, F_v is the field of v -adic numbers and in the infinite case, it is either $\mathbb{R}$ or $\mathbb{C}$. As a set $\mathbb{A}_F$ consists of all infinite tuples (x_v) where $x_v \in F_v$ and for all but finitely many places, x_v lies in the ring O_v of v -adic integers. We make $\mathbb{A}_F$ into a ring by defining addition and multiplication componentwise. Then we impose the adelic topology on it by declaring that for each finite set S of places of F which contains the infinite places, the subring

$$\prod_{v \in S} F_v \times \prod_{v \notin S} O_v,$$

endowed with the product topology, is a basic open neighbourhood of the identity. One then checks that with this topology, $\mathbb{A}_F$ is a locally compact topological ring. Moreover, one can think of F as being embedded in $\mathbb{A}_F$ via the map

$$x \mapsto (x, x, \dots).$$

Now let $n \geq 1$ and $G = GL_n$. For any commutative ring R with identity, $G(R)$ denotes the group of $n \times n$ matrices over R whose determinant is a unit of R . By M_r we denote a generic $r \times r$ matrix and by I_r the $r \times r$ identity matrix. The so-called standard parabolic subgroups of GL_n can be described as follows. They are in one-to-one correspondence with the partitions $n = n_1 + \dots + n_r$. For each such partition, the standard parabolic subgroup corresponding to this partition consists of matrices of the form

$$\begin{pmatrix} M_{n_1} & * & * \\ & \ddots & * \\ & & M_{n_r} \end{pmatrix},$$

and any parabolic subgroup is a GL_n conjugate of a standard parabolic subgroup. A parabolic subgroup has a Levi decomposition of the form $P = MN$ where N is the unipotent radical of P . M is called the Levi component of P . In the case of the standard parabolic, M and N can be described as consisting of matrices of the form

$$\begin{pmatrix} M_{n_1} & & \\ & \ddots & \\ & & M_{n_r} \end{pmatrix} \quad \begin{pmatrix} I_{n_1} & * & * \\ & \ddots & * \\ & & I_{n_r} \end{pmatrix}$$

respectively. We will write N_P to denote the unipotent radical in the Levi decomposition of a parabolic subgroup P .

The adelic topology on $G(\mathbb{A}_F)$ is similarly defined. Namely, for each finite set S as before, we declare

$$\prod_{v \in S} G(F_v) \times \prod_{v \notin S} G(O_v),$$

with the product topology, to be a basic open neighbourhood of the identity. The adelic topology on $G(\mathbb{A}_F)$ makes it into a locally compact group in which $G(F)$ is embedded diagonally as a discrete subgroup. If Z denotes the subgroup of scalar matrices in $G(\mathbb{A}_F)$, then the coset space $ZG(F) \backslash G(\mathbb{A}_F)$ with the quotient topology has finite volume with respect to any $G(\mathbb{A}_F)$ -invariant measure. This coset space is to be viewed as the higher dimensional analogue of the upper half-plane discussed earlier. We will now consider certain square integrable functions on this space.

5. Automorphic representations.

A character ω of $\mathbb{A}_F/F^*$ is called a Grossencharacter. We consider the Hilbert space

$$L^2(G(F) \backslash G(\mathbb{A}_F), \omega)$$

defined as the space of measurable functions ϕ on the coset space $G(F) \backslash G(\mathbb{A}_F)$ satisfying the conditions:

- (i) $\phi(zg) = \omega(z)\phi(g)$ for all $z \in Z$ and $g \in G(F) \backslash G(\mathbb{A}_F)$;
- (ii) $\int_{ZG(F) \backslash G(\mathbb{A}_F)} |\phi(g)|^2 dg < \infty$.

The subspace of cusp forms $L_0^2(G(F) \backslash G(\mathbb{A}_F), \omega)$ is defined by the extra condition

- (iii) for all parabolic groups P of $G(\mathbb{A}_F)$,

$$\int_{N_P(F) \backslash N_P(\mathbb{A}_F)} \phi(ng) dn = 0$$

for every $g \in G(\mathbb{A}_F)$.

Let R be the right regular representation of $G(\mathbb{A}_F)$ on $L^2(G(F) \backslash G(\mathbb{A}_F), \omega)$ given by the action

$$(R(g)\phi)(x) = \phi(xg)$$

for $\phi \in L^2(G(F) \backslash G(\mathbb{A}_F), \omega)$ and $x, g \in G(\mathbb{A}_F)$. This is a unitary representation of $G(\mathbb{A}_F)$. An automorphic representation is a subquotient of the right regular representation of $G(\mathbb{A}_F)$ on $L^2(G(F) \backslash G(\mathbb{A}_F), \omega)$. A cuspidal automorphic representation is a subrepresentation of the right regular representation of $G(\mathbb{A}_F)$ on $L_0^2(G(F) \backslash G(\mathbb{A}_F), \omega)$.

A representation of $G(\mathbb{A}_F)$ is called admissible if its restriction to the maximal compact subgroup

$$K = \prod_{v \text{ complex}} U_n(\mathbb{C}) \times \prod_{v \text{ real}} O_n(\mathbb{R}) \times \prod_{v \text{ finite}} GL_n(O_v)$$

contains each irreducible representation of K with only finite multiplicity.

To describe admissible representations, we need the notion of the restricted tensor product introduced in [JL]. Let $\{W_v : v \in V\}$ be a family of vector spaces indexed by V and V_0 a finite subset of V . In our case, V will be the set of places of F and V_0 the set of infinite places. For each $v \in V \setminus V_0$, let x_v be a non-zero vector in W_v . For each finite subset S of V containing V_0 , let

$$W_S = \otimes_{v \in S} W_v.$$

If $S \subset S'$, let

$$f_{S,S'} : W_S \rightarrow W_{S'}$$

be defined by

$$\otimes_{v \in S} w_v \mapsto \otimes_{v \in S} w_v \otimes_{v \in S' \setminus S} x_v.$$

Then, the restricted tensor product of the W_v with respect to the x_v , $W = \otimes_{x_v} W_v$ is defined as the inductive limit of the W_S . It is known (see [F, p. 181]) that any irreducible admissible representation of $GL_n(\mathbb{A}_F)$ can be written as a restricted tensor product $\otimes_v \pi_v$ where π_v is an irreducible representation of $GL_n(F_v)$. Moreover, the factors are unique up to equivalence.

For each finite v , one can construct certain representations of $GL_n(F_v)$ as follows. Consider the Borel subgroup

$$B(F_v) = \left\{ \begin{pmatrix} b_1 & * & * \\ & \ddots & * \\ & & b_n \end{pmatrix} \right\} \subseteq GL_n(F_v).$$

For any n -tuple of complex numbers, $z = (z_1, \dots, z_n)$, define χ_z on $B(F_v)$ by

$$\chi_z(b) = |b_1|_v^{z_1} \cdots |b_n|_v^{z_n},$$

where $|\cdot|_v$ denotes the v -adic norm. This gives us a quasi-character on $B(F_v)$. Let $\tilde{\pi}_{v,z}$ be the representation of $GL_n(F_v)$ obtained by inducing χ_z from $B(F_v)$ to $GL_n(F_v)$. We shall assume

$$\Re(z_1) \geq \Re(z_2) \geq \cdots \geq \Re(z_n).$$

Then, the Langlands classification theorem (see [BW, Chapter 11, §2]) tells us that $\tilde{\pi}_{v,z}$ has a unique irreducible quotient $\pi_{v,z}$. The representations obtained in this way are called the unramified principal series. If π_v is any representation of $GL_n(F_v)$ equivalent to $\pi_{v,z}$, we can define

$$A_v = \text{diag}(Nv^{-z_1}, \dots, Nv^{-z_n}) \in GL_n(\mathbb{C})$$

where Nv denotes the norm of v .

Now let $\pi = \otimes_v \pi_v$ be an irreducible, admissible automorphic representation of $GL_n(\mathbb{A}_F)$. As π is admissible, there is a finite set S of places containing the infinite places, such that π_v belongs to the unramified principal series. For such v , define

$$L_v(s, \pi_v) = \det(1 - A_v Nv^{-s})^{-1}$$

and set

$$L_S(s, \pi) = \prod_{v \notin S} L_v(s, \pi_v).$$

It is possible to define $L_v(s, \pi_v)$ for $v \in S$ so that the complete L -function

$$L(s, \pi) = \prod_v L_v(s, \pi_v)$$

extends to an entire function and satisfies a functional equation when π is cuspidal. This is a theorem of Godement and Jacquet [GJ].

The Ramanujan conjecture in this full generality predicts that all the eigenvalues of A_v are of absolute value 1 for cuspidal representations. This is still one of the central unresolved problems in the Langlands program. Below, we will discuss what is known in the general case.

In the case $n = 1$, an automorphic representation is a Grossencharacter and L -functions of Grossencharacters were studied in detail by Hecke [H] who established their analytic continuations and functional equations. Hecke used classical methods and this was reworked using the adelic framework by Tate in his celebrated thesis [Ta].

In the case $n = 2$ and $F = \mathbb{Q}$, the L -function of a cuspidal automorphic representation is either the L -function of a holomorphic cusp form (attached to some weight and level) or the L -function of a Maass wave form. Deligne's work [D] establishes the Ramanujan conjecture for only the holomorphic cusp forms. Thus the case of Maass forms is still left open.

In the case $n = 3$, Hoffstein and Murty [HM] have given a more classical treatment of L -functions attached to automorphic representations.

There are several expository accounts of the Langlands program treating the general case that we may refer the reader to. The first one is Gelbart's article [G] but this is already 20 years old. Another one is due to Murty [M2]. A more recent report is Knapp's article [Kn] and a progress report of Henniart [He].

The famous reciprocity conjecture of Langlands asserts that every Artin L -function $L(s, \rho; K/F)$ attached to an irreducible representation ρ of degree n must coincide with an $L(s, \pi)$ with π a cuspidal automorphic representation of $GL_n(\mathbb{A}_F)$. Since these latter L -functions have been shown to be entire, Artin's conjecture follows from the Langlands reciprocity conjecture.

As an extension of the modularity conjecture of Taniyama, Shimura and Weil, one expects that any reasonable L -function arising from arithmetic or a "motivic setting" should be the L -function of an automorphic representation. This is the general philosophy underlying the Langlands program.

6. Functoriality.

In the general set up, one considers $G(\mathbb{A}_F)$ with G an arbitrary reductive algebraic group instead of GL_n and thus, one can define L -functions attached automorphic representations in a similar way. One expects that no new objects arise in this fashion and somehow everything is covered by GL_n . This is perhaps a rather diluted way of expressing the idea of functoriality. A more precise way is to say that whenever there is a homomorphism between two reductive groups, there should be a corresponding map between the spaces of cuspidal automorphic representations. This would imply an identity of the corresponding L -functions. We will expand on this theme in the special case of GL_n . In this, we will consider only two aspects of the functoriality conjecture.

Let us denote by $\mathcal{A}_n(F)$ to be the space of automorphic representations of $GL_n(\mathbb{A}_F)$. For an arbitrary commutative ring R , we have the obvious map

$$GL_n(R) \times GL_m(R) \rightarrow GL_{nm}(R)$$

given by the tensor product of two matrices. This suggests that there should be a map

$$\mathcal{A}_n(F) \times \mathcal{A}_m(F) \rightarrow \mathcal{A}_{nm}(F)$$

such that given two automorphic representations π and π' , of degree n and m respectively, there should exist an automorphic representation Π in $\mathcal{A}_{nm}(F)$ so that

$$L(s, \Pi) = L(s, \pi \times \pi').$$

This conjecture makes two predictions. The first is the construction of an L -function $L(s, \pi \times \pi')$ from $L(s, \pi)$ and $L(s, \pi')$ and to establish its analytic continuation and functional equation. This has been done in a paper by Jacquet, Piatetskii-Shapiro and Shalika [JPS]. For $n = 2$, this is the classical work of Rankin [Ra] and Selberg [Sel2]. But the fact that it corresponds to an automorphic representation is conjectured by functoriality. For $n = 2$, this is a recent result of Ramakrishnan [Rm].

The symmetric power L -functions can be defined for any automorphic representation π in $\mathcal{A}_2(F)$ as follows. The simplest way to define them is as follows. As above, we have associated the matrix

$$A_v = \begin{pmatrix} \alpha_v & 0 \\ 0 & \beta_v \end{pmatrix}.$$

To this, we define the $(m+1) \times (m+1)$ matrix

$$A_v^{(m)} = \begin{pmatrix} \alpha_v^m & 0 & \cdots & 0 \\ 0 & \alpha_v^{m-1}\beta_v & \cdots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & \cdots & \beta_v^m \end{pmatrix}$$

and define the symmetric power L -function as

$$L(s, \text{Sym}^m(\pi)) = \prod_v L_v(s, \text{Sym}^m(\pi_v))$$

where for unramified v ,

$$L_v(s, \text{Sym}^m(\pi_v)) = \det(1 - A_v^{(m)} N v^{-s})^{-1}.$$

For the ramified places, a slight adjustment in the Euler factor is necessary. The main conjecture is that there should be an automorphic representation Π_m in $\mathcal{A}_{m+1}(F)$ such that

$$L(s, \Pi_m) = L(s, \text{Sym}^m(\pi))$$

so that we have a hypothesized “lifting” of automorphic forms on $GL(2)$ given by $\pi \mapsto \text{Sym}^m(\pi)$. In [La], Langlands showed that if we could prove that each of the L -functions $L(s, \text{Sym}^m(\pi))$ extends to an entire function and satisfies a functional equation, then the Sato-Tate conjecture, the Ramanujan conjecture and the Selberg eigenvalue conjecture follow immediately as a consequence. For the Sato-Tate conjecture, it is sufficient to have analytic continuation to $\Re(s) = 1$. For the Ramanujan and Selberg conjectures, the “weaker” assertion that they are analytic in some fixed half-plane is sufficient.

These L -functions, in the case that π corresponds to a holomorphic modular form coincide with the $L_m(s)$ defined in section 1. In 1978, Gelbart and Jacquet [GJa] established that symmetric square lift exists. That is, they proved that $\text{Sym}^2(\pi)$ is an automorphic representation of $\mathcal{A}_3(F)$, provided π is a non-monomial cuspidal representation. The recent work of Kim and Shahidi concerns the cases $m = 3$ and $m = 4$. It is now a theorem that the symmetric cube L -function is an entire function. For $m \leq 9$, we have meromorphic continuations of these functions. Below, we discuss the consequences of these results to the Selberg conjecture and the Ramanujan-Petersson conjecture. We summarise these results in:

Theorem 4. *The following maps exist:*

$\mathcal{A}_2(F)$	$\rightarrow$	$\mathcal{A}_3(F),$	$\pi \mapsto \text{Sym}^2(\pi);$	(Gelbart – Jacquet, 1978)
$\mathcal{A}_2(F)$	$\rightarrow$	$\mathcal{A}_4(F),$	$\pi \mapsto \text{Sym}^3(\pi);$	(Shahidi – Kim, 2000)
$\mathcal{A}_2(F)$	$\rightarrow$	$\mathcal{A}_5(F),$	$\pi \mapsto \text{Sym}^4(\pi)$	(Kim, 2001)
$\mathcal{A}_2(F) \times \mathcal{A}_2(F)$	$\rightarrow$	$\mathcal{A}_4(F),$	$(\pi, \pi') \mapsto \pi \times \pi'$	(Ramakrishnan, 2000) (Kim, 2001)
$\mathcal{A}_2(F) \times \mathcal{A}_3(F)$	$\rightarrow$	$\mathcal{A}_6(F),$	$(\pi, \pi') \mapsto \pi \times \pi';$	(Shahidi – Kim, 2000)

There is one more aspect of functoriality that is significant in the context of Artin’s conjecture. This has to do with base change and automorphic induction. Very briefly, this can be explained as follows.

Given an automorphic representation Π in $\mathcal{A}_n(K)$ where K is a finite extension of F , we may consider the L -function $L(s, \Pi)$. If we group together the Euler factors lying above a given place v of F , we obtain an Euler factor that looks like one of the local Langlands L -functions attached to an automorphic representation of F_v . This, together with the fact that Artin L -functions are invariant under induction suggests that there is an automorphic representation $\pi \in \mathcal{A}_{nd}(F)$ where $d = [K : F]$ so that

$$L(s, \Pi) = L(s, \pi).$$

One refers to π as the automorphic induction of Π .

A dual notion is the concept of base change. Given any automorphic representation π of $\mathcal{A}_n(F)$, one may twist the representation by Dirichlet characters corresponding to an abelian extension K/F and thus get

$$\prod_{\chi} L(s, \pi \otimes \chi).$$

That there is an automorphic representation $\Pi \in \mathcal{A}_n(K)$ so that the above product equals $L(s, \Pi)$ is a famous theorem of Arthur and Clozel [AC]. In the case of a general Galois extension, one can make a similar construction but it is unknown at present whether Π exists. If it does, Murty [MRM] showed that Artin's conjecture follows immediately for all solvable extensions.

In fact, more can be proved. Suppose K/F is a finite extension of degree d and we have the automorphic induction map

$$\mathcal{A}_1(K) \rightarrow \mathcal{A}_d(F).$$

Then both the Artin conjectures and Langlands reciprocity conjectures follow immediately.

To see this, we begin by writing an Artin L -function attached to an irreducible character χ of $\text{Gal}(K/F)$ as

$$L(s, \chi; K/F) = \prod_i L(s, \text{Ind}_{H_i}^G \psi_i; K/K^{H_i})^{n_i}$$

with ψ_i an abelian character of a proper subgroup H_i of G , as explained in section 3. By the Artin reciprocity law, we obtain a Grossencharacter Π_i in $\mathcal{A}_1(K^{H_i})$ so that

$$L(s, \text{Ind}_{H_i}^G \psi_i; K/K^{H_i}) = L(s, \Pi_i).$$

By automorphic induction, we can write

$$L(s, \Pi_i) = L(s, \pi_i)$$

for $\pi_i \in \mathcal{A}_{d_i}(F)$ with $d_i = [K^{H_i} : F]$. After decomposing each π_i into cuspidal automorphic representations, we obtain

$$L(s, \chi, K/F) = \prod_j L(s, \pi_j)^{e_j}$$

with the π_j distinct (up to isomorphism) cuspidal automorphic representations and $e_j \in \mathbb{Z}$. Now consider the ‘‘Rankin-Selberg’’ L -function of both sides. By irreducibility and the fact that an Artin L -function is analytic at $s = 1$ unless it corresponds to the trivial character (in which case it has a simple pole), we deduce that $L(s, \chi \otimes \bar{\chi}, K/F)$ has a simple pole at $s = 1$. On the other hand, the right hand side becomes a product of L -functions of the form

$L(s, \pi_i \times \pi_j^*)$ which by the results [JPS] has no pole unless π_j^* equals the contragredient of π_i in which case it has a simple pole. We deduce

$$1 = \sum_j e_j^2.$$

As the e_j 's are integers, we deduce there is only one factor on the right hand side in the above factorisation $L(s, \pi_1)$ (say). Thus, $e_1 = \pm 1$. If $e_1 = -1$, the Artin L -function will fail to have trivial zeros (and will have 'trivial poles') and so we conclude that $e_1 = 1$. This means

$$L(s, \chi, K/F) = L(s, \pi)$$

for some $\pi \in \mathcal{A}_n(F)$ with $n = \chi(1)$, which establishes both the Artin conjecture and Langlands reciprocity.

7. Converse theory.

There seem to be few techniques available to establish the conjectures foreshadowed by the principle of functoriality. At present, there is the method of the trace formula, initiated by Selberg [Sel], and subsequently developed to arbitrary reductive algebraic groups by Langlands and Arthur. This method we shall not discuss here. The second method is described as converse theory and begins with a famous theorem of Weil [W].

The underlying idea is the following. Suppose we are given a sequence of complex numbers $\{a_n\}$ and we would like to show that there is a cusp form f of weight k and level N such that $a_f(n) = a_n$. If such a form exists, then clearly the Dirichlet series

$$L(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

extends to an entire function and satisfies a functional equation as implied by classical Hecke theory. In addition, the "twisting principle" tells us that if χ is a Dirichlet character mod q with q coprime to N , then the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{a_n \chi(n)}{n^s}$$

also extends to an entire function and satisfies a functional equation which is precisely given in terms of Gauss sums, N , q and the Gamma function. The converse theorem of Weil says that these conditions are sufficient. A simplified proof of Weil's theorem was given later by Razar [Raz] who also reduced the infinite set of sufficient conditions of Weil to a finite set. Here is the precise form of Weil's theorem.

Theorem 5. (Weil) *Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of complex numbers satisfying $a_n = O(n^c)$ for some constant $c > 0$. Fix a positive integer N and an even positive integer k . Let $\epsilon = \pm 1$. Suppose that*

$$\Lambda(s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(s)$$

extends to an entire function, is bounded in vertical strips and satisfies a functional equation

$$\Lambda(s) = \epsilon(-1)^{k/2} \Lambda(k-s).$$

Suppose further that for every $(q, N) = 1$, and every primitive character $\chi \pmod{q}$, the series

$$L_\chi(s) = \sum_{n=1}^{\infty} \frac{a_n \chi(n)}{n^s}$$

is such that

$$\Lambda_\chi(s) = (q^2 N)^{s/2} (2\pi)^{-s} \Gamma(s) L_\chi(s)$$

extends to an entire function, is bounded in vertical strips and satisfies

$$\Lambda_\chi(s) = \epsilon(-1)^{k/2} \frac{g_\chi}{g_{\bar{\chi}}} \chi(-N) \Lambda_{\bar{\chi}}(k-s)$$

where g_χ is the Gauss sum

$$g_\chi = \sum_{a \bmod q} \chi(a) e^{2\pi i a/q}.$$

Suppose further that $L(s)$ converges absolutely at $s = k - \delta$ for some $\delta > 0$, then

$$f(z) = \sum_{n=1}^{\infty} a_n e^{2\pi i n z}$$

is a cusp form of weight k and level N .

This theorem was generalized by Jacquet and Langlands [JL] to the context of automorphic forms on $GL_2(\mathbb{A}_F)$ and recently to $GL_n(\mathbb{A}_F)$ by Cogdell and Piatetskii-Shapiro [CPS].

Converse theory has been a main tool in several of the striking achievements of the Langlands program thus far. For example, the Gelbart-Jacquet lift [GJ], the work of Shahidi and Kim [KS] as well as the recent work of Ramkrishnan [Rm] establishing the modularity of the Rankin-Selberg L -function, all use converse theory in a fundamental way.

8. Estimating Fourier coefficients.

An idea that goes back to Rankin [Ra] and Selberg [Sel2] is that if one has a sequence of complex numbers $\{a_n\}$ and the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{|a_n|^2}{n^s}$$

extends analytically to the entire complex plane apart from a simple pole at $s = 1$, and satisfies a functional equation, then, it is possible to estimate the coefficients a_n in a precise way. By methods of analytic number theory, one can derive a formula of the type

$$\sum_{n \leq x} |a_n|^2 = rx + O(x^\theta)$$

with some non-zero constant r given by the residue at $s = 1$ and a value of $\theta < 1$ determined by the growth of the function (usually guided by the gamma factors in the functional equation). By putting $x = N$ and then $x = N - 1$ with N an integer and taking differences leads to the estimate $a_n = O(n^{\theta/2})$. It was this kind of method that first led to the significant advance on the Ramanujan-Petersson conjecture.

The technique actually goes back to Landau [L]. He proved:

Theorem 6. (Landau) *Let $a_n \geq 0$ and set $f(s) = \sum_{n=1}^{\infty} a_n/n^s$. Suppose $f(s)$ is convergent in some half-plane and that it has analytic continuation for all $s \in \mathbb{C}$ except for a pole of order k at $s = 1$ and that it satisfies a functional equation*

$$Q^s \Delta(s) f(s) = Q^{1-s} \Delta(1-s) f(1-s)$$

where Q is a positive constant and

$$\Delta(s) = \prod_{i=1}^N \Gamma(\alpha_i s + \beta_i).$$

Set $d = 2 \sum_{i=1}^N \alpha_i$. Then,

$$\sum_{n \leq x} a_n = x P_{k-1}(\log x) + O(x^{\frac{d-1}{d+1}} \log^{k-1} x),$$

where P_{k-1} is a polynomial of degree $k - 1$.

An immediate corollary to this theorem is that $a_n = O(n^{\frac{d-1}{d+1}} \log^{k-1} n)$. Such a result when combined with the knowledge of analyticity of the various symmetric power L -functions leads to improved estimates that converge to the Ramanujan-Petersson conjecture and the Selberg eigenvalue conjecture. For instance, in the case $F = \mathbb{Q}$, and f is a Maass wave form, Kim and Shahidi [KSk] obtain $a_p(f) = O(p^{7/64})$. A similar estimate is derived for the Selberg conjecture. For the general case of an arbitrary number field F , one obtains an exponent of $1/9$ by the work of Shahidi.

This overview, by its brevity, omits discussion of other important results and contributors. For example, we did not discuss the triple product L -functions which are essential in the work of Ramakrishnan [Rm] and Kim [K]. We did not discuss the trace formula and the important work in this area. For these and other central features of the program, we direct the reader to the excellent expositions [Kn] and [He]. For the student who would like to have a more detailed instruction, we recommend [Bu] and [WL].

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